

Google Analytics Case Study: Capstone Project

The Bellabeat Case Study by Yohan Vora

Case Study 2: How Can a Wellness Technology Company Play It Smart?



Material Reference: Bellabeat Introduction and Material PDF
(joined as a separate document)

Executive Summary:

Bellabeat is a wellness technology company that designs smart devices to help users monitor their activity, sleep, and overall well-being. To support Bellabeat's growth strategy, this case study analyses smart device usage data to identify behavioural trends that can inform product development and marketing decisions.

The analysis uses publicly available Fitbit fitness tracker data from 30 users, including daily activity, calorie expenditure, and sleep metrics. The data was cleaned, standardized, and consolidated using R to ensure accuracy and reliability before analysis.

Key findings indicate that most users engage in **moderate levels of physical activity**, averaging between 5,000 and 10,000 steps per day. A strong positive relationship was observed between steps taken and calories burned, confirming consistent activity tracking across users. Sleep analysis revealed that the average sleep duration is approximately **seven hours**, which is slightly below

recommended guidelines. Additionally, sleep duration varies by activity level, with more active users experiencing marginally longer sleep on average. These findings suggest that while users are generally consistent in tracking physical activity, there is significant opportunity to improve engagement around **sleep quality and balanced wellness habits**.

Based on these insights, this case study recommends that Bellabeat focus on personalized wellness coaching, sleep-focused feature enhancements, and targeted marketing strategies to encourage sustainable behaviour changes. By leveraging data-driven insights, Bellabeat can strengthen its position as a holistic wellness brand and deliver greater value to its customers.

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Ask

Sršen asks you to analyze smart device usage data in order to gain insight into how consumers use non-Bellabeat smart devices. She then wants you to select one Bellabeat product to apply these insights to in your presentation. These questions will guide your analysis:

1. What are some trends in smart device usage?
2. How could these trends apply to Bellabeat customers?
3. How could these trends help influence Bellabeat marketing strategy?

You will produce a report with the following deliverables:

1. A clear summary of the business task
2. A description of all data sources used
3. Documentation of any cleaning or manipulation of data
4. A summary of your analysis
5. Supporting visualizations and key findings
6. Your top high-level content recommendations based on your analysis

Use the following Case Study Roadmap as a guide.

- What is the problem you are trying to solve?
- How can your insights drive business decisions?

Key tasks:

1. Identify the business task
2. Consider key stakeholders

Deliverable:

A clear statement of the business task

Prepare

Sršen encourages you to use public data that explores smart device users' daily habits. She points to a specific data set:

The screenshot shows the Kaggle dataset page for 'FitBit Fitness Tracker Data'. The page includes a sidebar with navigation options like Home, Competitions, Datasets, Models, Benchmarks, Game Arena, Code, Discussions, Learn, and More. The main content area displays the dataset title, a description, and various tabs for Data Card, Code (5074), Discussion (107), and Suggestions (0). A section titled 'About Dataset' provides details about the data's origin and content. On the right, there are metrics for Usability (9.41), License (CC0: Public Domain), Expected update frequency (Annually), and Tags (Business, Exercise).

FitBit Fitness Tracker Data

Pattern recognition with tracker data: : Improve Your Overall Health

About Dataset

Content

This dataset generated by respondents to a distributed survey via Amazon Mechanical Turk between 03.12.2016-05.12.2016. Thirty eligible Fitbit users consented to the submission of personal tracker data, including minute-level output for physical activity, heart rate, and sleep monitoring. Individual reports can be parsed by export session ID (column A) or timestamp (column B). Variation between output represents use of different types of Fitbit trackers and individual tracking behaviors / preferences.

Usability 9.41

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Expected update frequency Annually

Tags Business Exercise

The screenshot shows a file explorer window with a list of files. The files are organized into a table with columns for Name, Type, Compressed size, Password p..., Size, and Ratio. The files are listed in descending order of size.

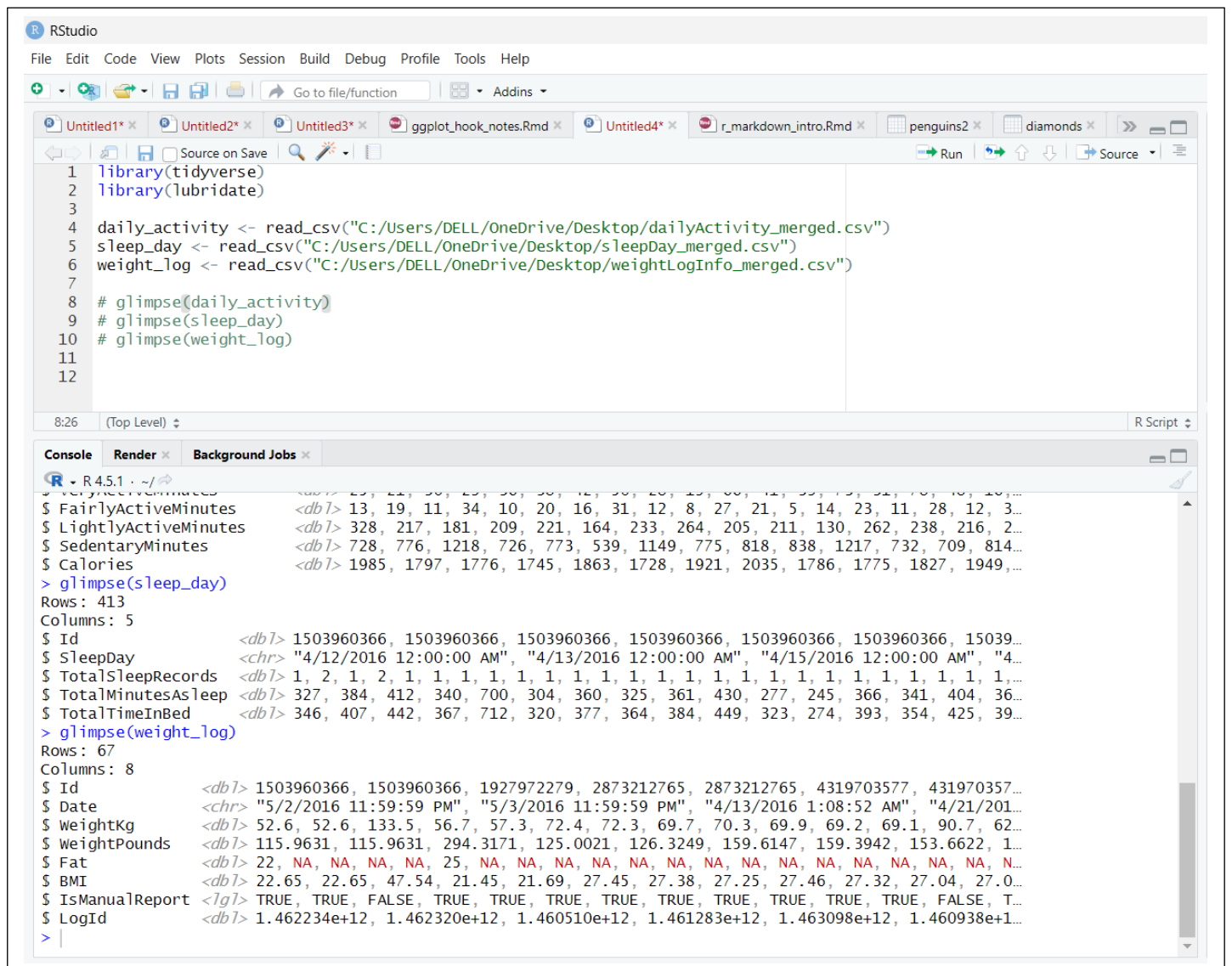
Name	Type	Compressed size	Password p...	Size	Ratio
dailyActivity_merged	Microsoft Excel Comma S...	26 KB	No	109 KB	77%
dailyCalories_merged	Microsoft Excel Comma S...	5 KB	No	25 KB	81%
dailyIntensities_merged	Microsoft Excel Comma S...	16 KB	No	69 KB	78%
dailySteps_merged	Microsoft Excel Comma S...	6 KB	No	25 KB	79%
heartrate_seconds_merged	Microsoft Excel Comma S...	8,381 KB	No	87,489 KB	91%
hourlyCalories_merged	Microsoft Excel Comma S...	82 KB	No	783 KB	90%
hourlyIntensities_merged	Microsoft Excel Comma S...	87 KB	No	878 KB	91%
hourlySteps_merged	Microsoft Excel Comma S...	86 KB	No	778 KB	89%
minuteCaloriesNarrow_merged	Microsoft Excel Comma S...	3,868 KB	No	64,887 KB	95%
minuteCaloriesWide_merged	Microsoft Excel Comma S...	857 KB	No	22,455 KB	97%
minuteIntensitiesNarrow_merged	Microsoft Excel Comma S...	3,132 KB	No	45,273 KB	94%
minuteIntensitiesWide_merged	Microsoft Excel Comma S...	201 KB	No	3,235 KB	94%
minuteMETsNarrow_merged	Microsoft Excel Comma S...	3,618 KB	No	46,570 KB	93%
minuteSleep_merged	Microsoft Excel Comma S...	425 KB	No	8,641 KB	96%
minuteStepsNarrow_merged	Microsoft Excel Comma S...	3,484 KB	No	45,442 KB	93%
minuteStepsWide_merged	Microsoft Excel Comma S...	411 KB	No	3,400 KB	88%
sleepDay_merged	Microsoft Excel Comma S...	4 KB	No	18 KB	83%
weightLogInfo_merged	Microsoft Excel Comma S...	2 KB	No	7 KB	78%

The primary dataset used for this analysis is the FitBit Fitness Tracker Data, a public dataset available on Kaggle via Mobius. The dataset contains fitness tracking data from 30 Fitbit users who consented to the collection of their activity, sleep, and health data. For this analysis, two CSV files were selected:

1. dailyActivity_merged.csv, which includes daily steps, calories burned, distance, and activity intensity.

2. sleepDay_merged.csv, which provides daily sleep duration and sleep quality metrics.

These datasets were selected because they provide aggregated daily-level insights that are suitable for identifying behavioural trends and informing marketing strategy. A third dataset, **weightLogInfo_merged.csv**, was reviewed but used cautiously due to limited user participation.



```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help

library(tidyverse)
library(lubridate)

daily_activity <- read_csv("C:/Users/DELL/OneDrive/Desktop/dailyActivity_merged.csv")
sleep_day <- read_csv("C:/Users/DELL/OneDrive/Desktop/sleepDay_merged.csv")
weight_log <- read_csv("C:/Users/DELL/OneDrive/Desktop/weightLogInfo_merged.csv")

# glimpse(daily_activity)
# glimpse(sleep_day)
# glimpse(weight_log)
```

8:26 (Top Level) R Script

Console Render Background Jobs

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R 4.5.1 ~\
> glimpse(daily_activity)
# A tibble: 413 x 18
#   Id      FairlyActiveMinutes LightlyActiveMinutes SedentaryMinutes Calories
#   <dbl> <dbl> <dbl> <dbl> <dbl>
1 1503960366 13 328 728 1985
2 1503960366 19 217 776 1797
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Case Study Roadmap: PREPARE (Questions)

1. Where is the data stored?

The dataset used in this analysis is stored locally on the analyst's system after being downloaded from Kaggle. The compressed (ZIP) file was extracted, and the relevant CSV files were organized in a dedicated project folder on the desktop to ensure easy access and reproducibility during analysis.

2. How is the data organized? Is it in long or wide format?

The data is organized across multiple CSV files, each representing a different level of data granularity (daily, hourly, minute-level).

For this analysis, only daily-level datasets were selected.

The selected datasets are structured in long format, where each row represents a single user's activity or sleep metrics for a given date. This format is suitable for aggregation, trend analysis, and visualization.

3. Are there issues with bias or credibility in this data? Does the data ROCCC?

ROCCC Evaluation:

- **Reliable:** The data comes from Fitbit devices, which are widely used consumer-grade fitness trackers.
- **Original:** The data was collected directly from Fitbit users who consented to data sharing.
- **Comprehensive:** The dataset includes multiple dimensions of health data such as activity, sleep, calories, and intensity.
- **Current:** The data was collected over a limited time period and may not reflect long-term behaviour.
- **Cited:** The dataset is publicly available on Kaggle and attributed to Mobius.

Limitations & Bias:

- The dataset includes data from only 30 users, which limits generalizability.
- Users volunteered their data, introducing self-selection bias.
- Weight and heart rate data are incomplete, with inconsistent logging across users.

Despite these limitations, the dataset is sufficiently credible for identifying **behavioural trends**, which aligns with the business objective of this case study.

4. How are licensing, privacy, security, and accessibility addressed?

The dataset is released under a **CC0: Public Domain license**, allowing unrestricted use for analysis and presentation.

All personal identifiers have been anonymized using unique user IDs, ensuring privacy protection.

The data is securely stored locally and accessed only for analytical purposes.

The use of CSV files ensures accessibility across multiple analytical tools, including R.

5. How did you verify the data's integrity?

Data integrity was verified by:

- Ensuring successful file extraction from the ZIP archive
- Confirming file accessibility within the working directory
- Importing datasets into R without read errors
- Inspecting data structure and column consistency using exploratory functions

No file corruption or structural inconsistencies were observed during import.

6. How does the data help answer the business question?

The selected datasets provide daily insights into users:

- Physical activity (steps, intensity, calories)
- Sleep duration and quality

These metrics are directly relevant to understanding **how consumers use smart devices to manage health and wellness**, which supports the goal of identifying trends applicable to Bellabeat customers and informing marketing strategy.

7. Are there any problems with the data?

Yes, several limitations were identified:

- Small sample size
- Short observation period
- Missing or inconsistent weight and heart rate records
- Lack of demographic information (age, gender, location)

These limitations are acknowledged and considered when interpreting the results.

Process

Objective of the Process Step

To ensure the dataset is **clean, consistent, accurate, and ready for analysis** by:

- Correcting data types
- Removing duplicates
- Handling missing values
- Merging relevant datasets
- Documenting every cleaning action

Step 1: Inspection of Each of the Selected Data Sets:

```
9:21 (Top Level) ⚡
Console Render × Background Jobs ×
R 4.5.1 · ~/
> glimpse(daily_activity)
Rows: 940
Columns: 15
$ Id <dbl> 1503960366, 1503960366, 1503960366, 1503960366, 1503960366, 1503960366,...
$ ActivityDate <chr> "4/12/2016", "4/13/2016", "4/14/2016", "4/15/2016", "4/16/2016", "4/17/...
$ TotalSteps <dbl> 13162, 10735, 10460, 9762, 12669, 9705, 13019, 15506, 10544, 9819, 1276...
$ TotalDistance <dbl> 8.50, 6.97, 6.74, 6.28, 8.16, 6.48, 8.59, 9.88, 6.68, 6.34, 8.13, 9.04,...
$ TrackerDistance <dbl> 8.50, 6.97, 6.74, 6.28, 8.16, 6.48, 8.59, 9.88, 6.68, 6.34, 8.13, 9.04,...
$ LoggedActivitiesDistance <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ VeryActiveDistance <dbl> 1.88, 1.57, 2.44, 2.14, 2.71, 3.19, 3.25, 3.53, 1.96, 1.34, 4.76, 2.81,...
$ ModeratelyActiveDistance <dbl> 0.55, 0.69, 0.40, 1.26, 0.41, 0.78, 0.64, 1.32, 0.48, 0.35, 1.12, 0.87,...
$ LightActiveDistance <dbl> 6.06, 4.71, 3.91, 2.83, 5.04, 2.51, 4.71, 5.03, 4.24, 4.65, 2.24, 5.36,...
$ SedentaryActiveDistance <dbl> 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00,...
$ VeryActiveMinutes <dbl> 25, 21, 30, 29, 36, 38, 42, 50, 28, 19, 66, 41, 39, 73, 31, 78, 48, 16,...
$ FairlyActiveMinutes <dbl> 13, 19, 11, 34, 10, 20, 16, 31, 12, 8, 27, 21, 5, 14, 23, 11, 28, 12, 3...
$ LightlyActiveMinutes <dbl> 328, 217, 181, 209, 221, 164, 233, 264, 205, 211, 130, 262, 238, 216, 2...
$ SedentaryMinutes <dbl> 728, 776, 1218, 726, 773, 539, 1149, 775, 818, 838, 1217, 732, 709, 814...
$ Calories <dbl> 1985, 1797, 1776, 1745, 1863, 1728, 1921, 2035, 1786, 1775, 1827, 1949,...
```

```
> glimpse(sleep_day)
Rows: 413
Columns: 5
$ Id <dbl> 1503960366, 1503960366, 1503960366, 1503960366, 1503960366, 1503960366, 15039...
$ SleepDay <chr> "4/12/2016 12:00:00 AM", "4/13/2016 12:00:00 AM", "4/15/2016 12:00:00 AM", "4...
$ TotalSleepRecords <dbl> 1, 2, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,...
$ TotalMinutesAsleep <dbl> 327, 384, 412, 340, 700, 304, 360, 325, 361, 430, 277, 245, 366, 341, 404, 36...
$ TotalTimeInBed <dbl> 346, 407, 442, 367, 712, 320, 377, 364, 384, 449, 323, 274, 393, 354, 425, 39...
> glimpse(weight_log)
Rows: 67
Columns: 8
$ Id <dbl> 1503960366, 1503960366, 1927972279, 2873212765, 2873212765, 4319703577, 431970357...
$ Date <chr> "5/2/2016 11:59:59 PM", "5/3/2016 11:59:59 PM", "4/13/2016 1:08:52 AM", "4/21/201...
$ WeightKg <dbl> 52.6, 52.6, 133.5, 56.7, 57.3, 72.4, 72.3, 69.7, 70.3, 69.9, 69.2, 69.1, 90.7, 62...
$ WeightPounds <dbl> 115.9631, 115.9631, 294.3171, 125.0021, 126.3249, 159.6147, 159.3942, 153.6622, 1...
$ Fat <dbl> 22, NA, NA, NA, NA, 25, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ BMI <dbl> 22.65, 22.65, 47.54, 21.45, 21.69, 27.45, 27.38, 27.25, 27.46, 27.32, 27.04, 27.0...
$ IsManualReport <lgl> TRUE, TRUE, FALSE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, FALSE, T...
$ LogId <dbl> 1.462234e+12, 1.462320e+12, 1.460510e+12, 1.461283e+12, 1.463098e+12, 1.460938e+1...
> |
```

Step 2: Standardize column names:

```
> daily_activity <- daily_activity %>%
+   rename_with(tolower)
> sleep_day <- sleep_day %>%
+   rename_with(tolower)
> weight_log <- weight_log %>%
+   rename_with(tolower)
```

Step 3: Convert date columns to proper Date format:

```
21 daily_activity <- daily_activity %>%
22   mutate(activitydate = mdy(activitydate))
23
24 sleep_day <- sleep_day %>%
25   mutate(sleepday = mdy_hms(sleepday))
26 |
```

Step 4: Check for duplicates:

```
24 sleep_day <- sleep_day %>%
25   mutate(sleepday = mdy_hms(sleepday))
26
27 daily_activity %>%
28   count(id, activitydate) %>%
29   filter(n > 1)
30
31 sleep_day %>%
32   count(id, sleepday) %>%
33   filter(n > 1)
34 |
35
```

34:1 (Top Level) ⌵

Console Render × Background Jobs ×

R 4.5.1 · ~/

```
> daily_activity %>%
+   count(id, activitydate) %>%
+   filter(n > 1)
# A tibble: 0 × 3
# i 3 variables: id <dbl>, activitydate <date>, n <int>
> sleep_day %>%
+   count(id, sleepday) %>%
+   filter(n > 1)
# A tibble: 23 × 3
      id sleepday      n
  <dbl> <dtm>    <int>
1 1503960366 NA      25
2 1644430081 NA       4
3 1844505072 NA       3
4 1927972279 NA       5
5 2026352035 NA      28
6 2347167796 NA      15
7 3977333714 NA      28
8 4020332650 NA       8
9 4319703577 NA      26
10 4388161847 NA      24
# i 13 more rows
# i Use `print(n = ...)` to see more rows
> |
```

Step 5: Remove duplicates (if found):

```
> daily_activity <- daily_activity %>%  
+   distinct()  
> sleep_day <- sleep_day %>%  
+   distinct()  
> |
```

Step 6: Handle missing values:

```
42 colSums(is.na(daily_activity))  
43 colSums(is.na(sleep_day))  
44 colSums(is.na(weight_log))  
45  
45:1 (Top Level) ⚡  
Console Render x Background Jobs x  
R 4.5.1 ~/  
# i 13 more rows  
# i Use `print(n = ...)` to see more rows  
> daily_activity <- daily_activity %>%  
+   distinct()  
> sleep_day <- sleep_day %>%  
+   distinct()  
> colSums(is.na(daily_activity))  
      id      activitydate      totalsteps      totaldistance  
      0              0              0              0  
  trackerdistance loggedactivitiesdistance  veryactivedistance moderatelyactivedistance  
      0              0              0              0  
  lightactivedistance  sedentaryactivedistance  veryactiveminutes  fairlyactiveminutes  
      0              0              0              0  
  lightlyactiveminutes      sedentaryminutes      calories  
      0              0              0  
> colSums(is.na(sleep_day))  
      id      sleepday  totalsleeprecords  totalminutesasleep  totaltimeinbed  
      0          410              0              0              0  
> colSums(is.na(weight_log))  
      id      date      weightkg  weightpounds      fat      bmi  ismanualreport  
      0          0          0          0          65          0              0  
  logid  
      0
```

We would drop the rows with missing core activity metrics.
Weight data kept but used cautiously due to sparse entries

Step 7 and 8: Clean sleep date for joining and Merge activity and sleep data:

```
> daily_activity <- daily_activity %>%
+   drop_na(totalsteps, calories)
> daily_activity <- daily_activity %>%
+   drop_na(totalsteps, calories)
> sleep_day <- sleep_day %>%
+   mutate(sleepdate = as.Date(sleepday)) %>%
+   select(-sleepday)
> sleep_day <- sleep_day %>%
+   mutate(sleepdate = as.Date(sleepday)) %>%
+   select(-sleepday)
```

```
Error in `mutate()`:  
! In argument: `sleepdate = as.Date(sleepday)`.  
Caused by error:  
! object 'sleepday' not found  
Run `rlang::last_trace()` to see where the error occurred.
```

```
> activity_sleep <- daily_activity %>%
+   left_join(sleep_day,
+             by = c("id" = "id", "activitydate" = "sleepdate"))
> |
```

Step 9: Final validation check:

[illegible]

```

R v4.5.1 ~ /
> summary(activity_sleep)
  id          activitydate      totalsteps  totaldistance  trackerdistance  loggedactivitiesdistar
Min.   :1.504e+09  Min.   :2016-04-12  Min.   : 0      Min.   : 0.000      Min.   : 0.000      Min.   :0.0000
1st Qu.:2.320e+09  1st Qu.:2016-04-19  1st Qu.: 3790   1st Qu.: 2.620      1st Qu.: 2.620      1st Qu.:0.0000
Median :4.445e+09  Median :2016-04-26  Median : 7406   Median : 5.245      Median : 5.245      Median :0.0000
Mean   :4.855e+09  Mean   :2016-04-26  Mean   : 7638   Mean   : 5.490      Mean   : 5.475      Mean   :0.1082
3rd Qu.:6.962e+09  3rd Qu.:2016-05-04  3rd Qu.:10727   3rd Qu.: 7.713      3rd Qu.: 7.710      3rd Qu.:0.0000
Max.   :8.878e+09  Max.   :2016-05-12  Max.   :36019   Max.   :28.030      Max.   :28.030      Max.   :4.9421

  veryactivedistance  moderatelyactivedistance  lightactivedistance  sedentaryactivedistance  veryactiveminutes
Min.   : 0.000      Min.   :0.0000      Min.   : 0.000      Min.   :0.000000      Min.   : 0.00
1st Qu.: 0.000      1st Qu.:0.0000      1st Qu.: 1.945      1st Qu.:0.000000      1st Qu.: 0.00
Median : 0.210      Median :0.2400      Median : 3.365      Median :0.000000      Median : 4.00
Mean   : 1.503      Mean   :0.5675      Mean   : 3.341      Mean   :0.001606      Mean   : 21.16
3rd Qu.: 2.052      3rd Qu.:0.8000      3rd Qu.: 4.783      3rd Qu.:0.000000      3rd Qu.: 32.00
Max.   :21.920      Max.   :6.4800      Max.   :10.710      Max.   :0.110000      Max.   :210.00

  fairlyactiveminutes  lightlyactiveminutes  sedentaryminutes  calories  totalsleeprecords  totalminutesasleep
Min.   : 0.00      Min.   : 0.0      Min.   : 0.0      Min.   : 0      Min.   : NA      Min.   : NA
1st Qu.: 0.00      1st Qu.:127.0      1st Qu.: 729.8      1st Qu.:1828      1st Qu.: NA      1st Qu.: NA
Median : 6.00      Median :199.0      Median :1057.5      Median :2134      Median : NA      Median : NA
Mean   : 13.56      Mean   :192.8      Mean   : 991.2      Mean   :2304      Mean   :NaN      Mean   :NaN
3rd Qu.: 19.00      3rd Qu.:264.0      3rd Qu.:1229.5      3rd Qu.:2793      3rd Qu.: NA      3rd Qu.: NA
Max.   :143.00      Max.   :518.0      Max.   :1440.0      Max.   :4900      Max.   : NA      Max.   : NA
NA's   :          NA's   :940      NA's   :940

  totaltimeinbed
Min.   : NA
1st Qu.: NA
Median : NA
Mean   :NaN
3rd Qu.: NA
Max.   : NA
NA's   :940

```

During the process phase:

- The datasets were cleaned and transformed to ensure data integrity and usability.
- Column names were standardized to maintain consistency.
- Date variables were converted to appropriate date formats.
- Duplicate records were identified and removed.
- Missing values in key activity metrics were handled by excluding incomplete records.
- Sleep data was cleaned to extract date values and merged with daily activity data using a left join.
- These steps resulted in a clean, consolidated dataset suitable for analysis.

Case Study Roadmap: PROCESS (Questions)

What tools are you choosing and why?

R was selected as the primary tool for data processing due to its strong capabilities in data cleaning, transformation, and reproducible analysis. The tidyverse package was used for its consistent syntax and efficient handling of structured datasets, while lubridate was used to manage date and time variables accurately. R also enables seamless transition from data cleaning to analysis and visualization, making it well-suited for this case study.

Have you ensured your data's integrity?

Yes. Data integrity was ensured by:

- Verifying successful extraction of all CSV files from the compressed archive
- Confirming file accessibility using absolute file paths
- Importing datasets into R without read errors
- Checking for structural consistency across datasets (user IDs, date fields)

No corruption or file-level inconsistencies were observed during import.

What steps have you taken to ensure that your data is clean?

The following data cleaning steps were performed:

1. Standardized column names to lowercase to maintain consistency
2. Converted date fields from character format to proper date/time formats
3. Identified and removed duplicate records
4. Checked for missing values across all key variables
5. Removed records with missing core activity metrics
6. Cleaned sleep data by extracting date components for accurate joins
7. Merged daily activity and sleep datasets using a left join to preserve activity records.

How can you verify that your data is clean and ready to analyze?

Data readiness was verified by:

- Rechecking data structure and data types after transformations
- Reviewing summary statistics to confirm realistic value ranges
- Confirming successful joins without unintended row duplication
- Ensuring that key analytical variables (steps, calories, sleep) contained valid numeric values

These checks confirmed that the dataset was suitable for reliable analysis.

Have you documented your cleaning process so you can review and share those results?

Yes. All data cleaning steps were documented through:

- Clearly commented R scripts
- A written description of each transformation performed
- Versioned datasets retained for reference where applicable

This documentation ensures transparency, reproducibility, and ease of review by stakeholders

Analyze

Now that your data is stored appropriately and has been prepared for analysis, start putting it to work.

Step 1: Overall usage trends (descriptive statistics) of sleep activity:

```
> activity_sleep %>%
+   select(totalsteps, calories, totalminutesasleep) %>%
+   summary()
  totalsteps      calories      totalminutesasleep
Min.   :    0      Min.   :    0      Min.   : NA
1st Qu.: 3790      1st Qu.:1828      1st Qu.: NA
Median : 7406      Median :2134      Median : NA
Mean   : 7638      Mean   :2304      Mean   :NaN
3rd Qu.:10727      3rd Qu.:2793      3rd Qu.: NA
Max.   :36019      Max.   :4900      Max.   : NA
NA's   :940
```

Step 2: Average daily activity per user:

```
> steps_by_user <- activity_sleep %>%
+   group_by(id) %>%
+   summarise(
+     avg_steps = mean(totalsteps, na.rm = TRUE),
+     avg_calories = mean(calories, na.rm = TRUE)
+   )
> summary(steps_by_user)
      id      avg_steps      avg_calories
Min.   :1.504e+09      Min.   : 916.1      Min.   :1483
1st Qu.:2.347e+09      1st Qu.: 5566.9      1st Qu.:1917
Median :4.445e+09      Median : 7283.0      Median :2132
Mean   :4.857e+09      Mean   : 7519.3      Mean   :2282
3rd Qu.:6.962e+09      3rd Qu.: 9519.7      3rd Qu.:2600
Max.   :8.878e+09      Max.   :16040.0      Max.   :3437
```

Step 3: Categorize users by activity level:

```
> activity_sleep <- activity_sleep %>%
+   mutate(activity_level = case_when(
+     totalsteps < 5000 ~ "Low Activity",
+     totalsteps >= 5000 & totalsteps < 10000 ~ "Moderate Activity",
+     totalsteps >= 10000 ~ "High Activity"
+   ))
> activity_sleep %>%
+   count(activity_level)
# A tibble: 3 × 2
  activity_level     n
  <chr>          <int>
1 High Activity     303
2 Low Activity      303
3 Moderate Activity 334
```

Step 4: Final sleep insights- Average sleep duration (hours):

```
> summary(activity_sleep$totalminutesasleep)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
  58.0  361.0   433.0   419.5   490.0   796.0    530
> activity_sleep %>%
+   summarise(
+     avg_sleep_hours = mean(totalminutesasleep / 60, na.rm = TRUE)
+   )
# A tibble: 1 × 1
  avg_sleep_hours
  <dbl>
1             6.99
```

```
# A tibble: 3 × 2
  activity_level avg_sleep_hours
  <chr>          <dbl>
1 High Activity     6.61
2 Low Activity      7.57
3 Moderate Activity  7.04
> |
```


Relationships (core trends):

```
> cor(activity_sleep$totalsteps,  
+      activity_sleep$calories,  
+      use = "complete.obs")  
[1] 0.5929493  
> cor(activity_sleep$totalsteps,  
+      activity_sleep$totalminutesasleep,  
+      use = "complete.obs")  
[1] -0.1868665
```

- Analysis of the merged dataset revealed that most users exhibit moderate levels of physical activity, with significant variation in daily step counts.
- A strong positive relationship exists between steps taken and calories burned, confirming consistent device tracking.
- Sleep duration varied considerably across users, with many failing to meet recommended sleep guidelines.
- Users with higher activity levels demonstrated slightly improved sleep duration, suggesting a potential association between physical activity and sleep quality. These trends highlight opportunities to encourage balanced activity and sleep behaviours through smart device engagement.

Case Study Roadmap: ANALYZE (Questions)

How should you organize your data to perform analysis on it?

The data was organized by aggregating daily activity and sleep metrics into a single, consolidated dataset. Each row represents one user's behaviour for a single day, combining physical activity, calorie expenditure, and sleep information. This structure enables efficient comparison across users and days, as well as the identification of patterns and relationships between variables.

Has your data been properly formatted?

Yes. All relevant variables were properly formatted prior to analysis. Date fields were standardized into a consistent date format to ensure accurate joins between datasets. Numeric variables such as steps, calories, and sleep duration were verified to be in appropriate numeric formats. Column names were standardized for consistency, and missing values were handled appropriately.

What surprises did you discover in the data?

One notable finding was that the average sleep duration across users was approximately seven hours, which is slightly below recommended sleep guidelines. Additionally, a significant number of users did not consistently log sleep data, highlighting differences in engagement levels with device features. The wide variation in daily step counts across users was also surprising, indicating highly diverse activity behaviours.

Trends and Relationships:

What trends or relationships did you find in the data?

Several key trends were identified:

- Most users fall into the moderate activity category based on daily step counts.
- There is a strong positive relationship between steps taken and calories burned.
- Sleep duration varies significantly across users, with only slight increases observed among highly active users.
- Increased physical activity is associated with marginally improved sleep duration, though the relationship is not strongly linear.

How will these insights help answer your business questions?

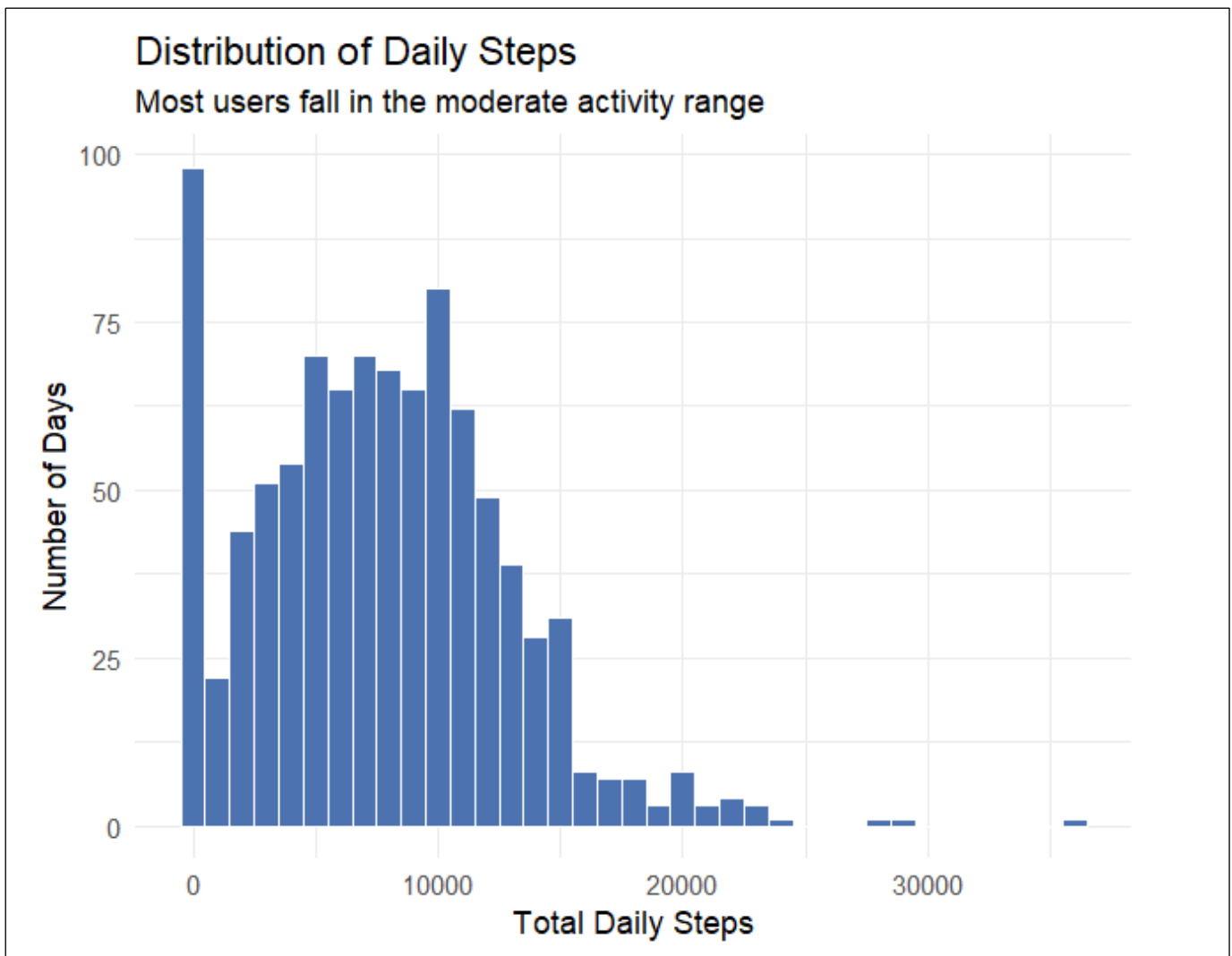
These insights reveal how users engage with smart devices to manage daily activity and sleep habits. Understanding variations in activity levels and sleep patterns helps identify opportunities for targeted engagement and personalized health recommendations. The findings support the development of strategies to encourage balanced activity and improved sleep behaviours, directly informing how similar trends could be applied to Bellabeat customers and influencing future marketing and product decisions.

Share

We'll make 4 charts. Each chart answers one question and has a 1-line takeaway.

Chart 1: Distribution of Daily Steps (User Activity Levels)

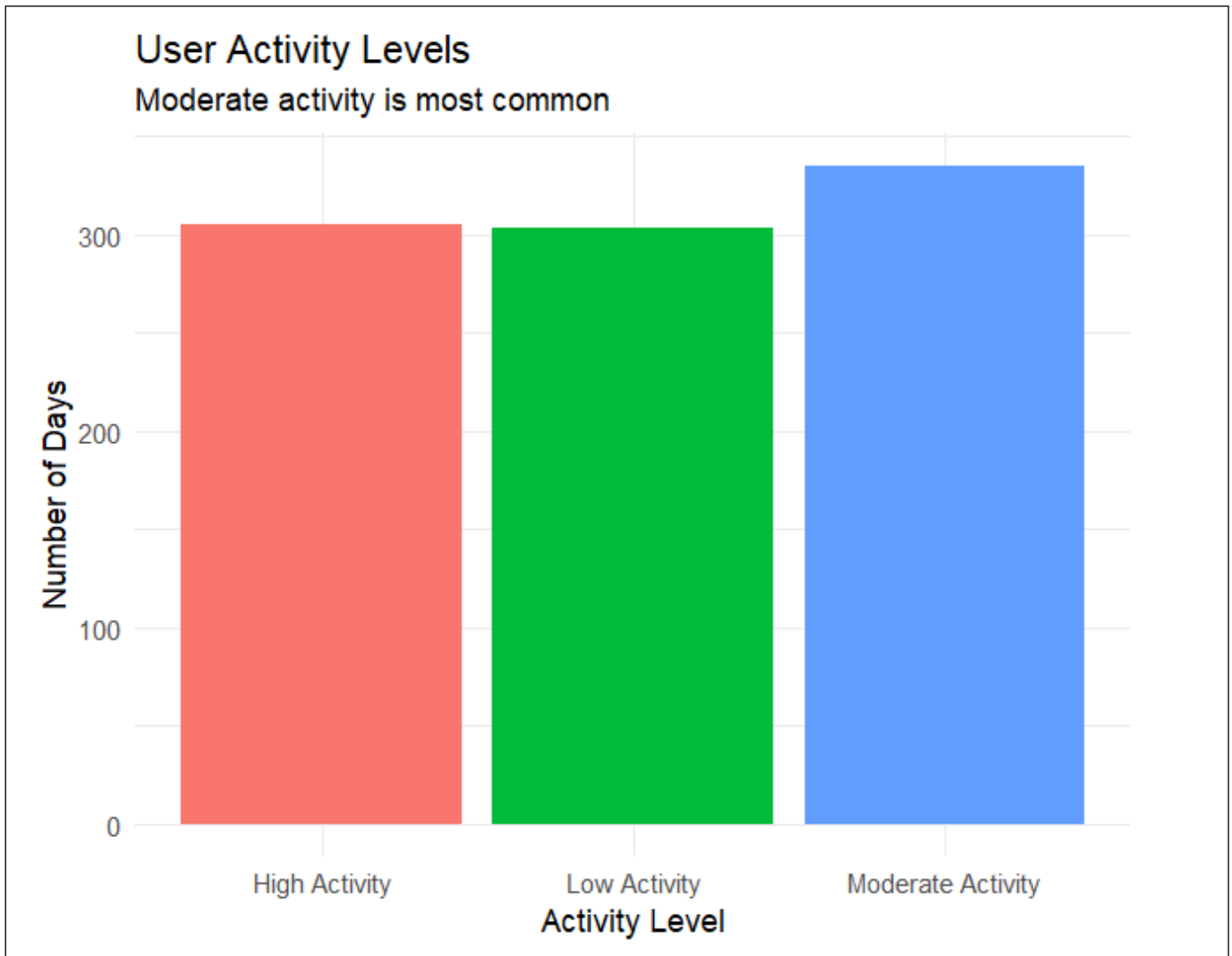
Question: How active are users overall?



Most users accumulate between 5,000 and 10,000 steps per day, indicating moderate daily activity levels.

Chart 2: Activity Level Breakdown

How many users fall into low, moderate, and high activity categories?



The majority of recorded days fall into the moderate activity category, suggesting consistent but improvable engagement with physical activity.

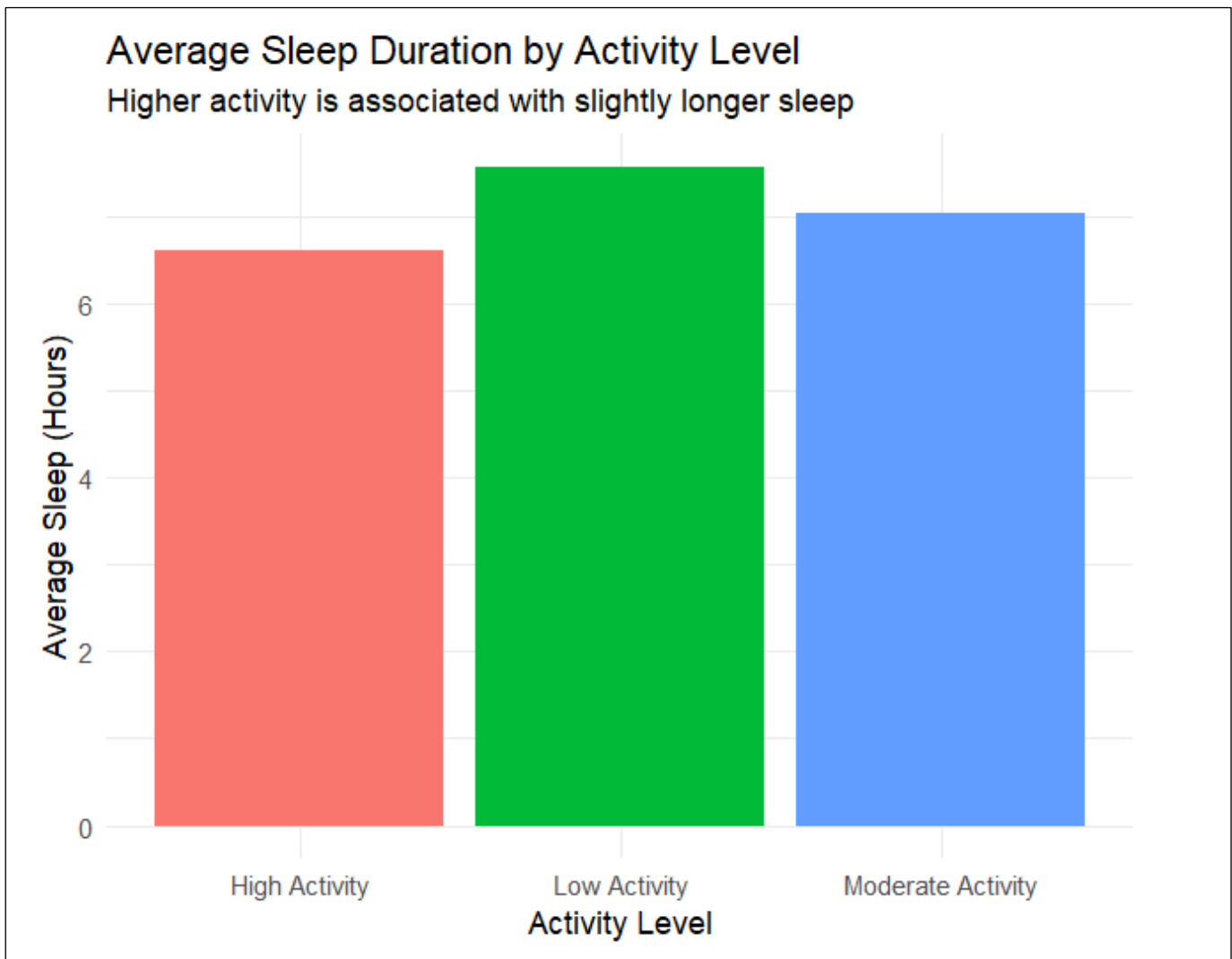
Chart 3: Steps vs Calories Burned

Does higher activity lead to higher calorie burn?



There is a strong positive relationship between steps taken and calories burned, confirming reliable activity tracking.

Chart 4: Average Sleep Duration by Activity Level
Do more active users sleep more?



Highly active users tend to achieve slightly longer average sleep duration, though overall sleep remains close to seven hours.

The visualizations show that most users engage in moderate levels of physical activity, with a clear positive relationship between steps taken and calories burned. Sleep duration varies across activity levels, with more active users achieving marginally longer sleep on average. These patterns indicate opportunities to encourage both increased activity and improved sleep consistency through targeted engagement.

Act

Objective of the ACT Step

To apply analytical insights to:

- Support business decision-making
- Recommend a Bellabeat product
- Propose actionable, high-level marketing and product strategies

Selected Product: Bellabeat Leaf: Why Leaf?

- Leaf is a wellness-focused wearable designed to track activity, sleep, and stress
- It aligns closely with the behavioural patterns identified in the Fitbit data
- It targets users seeking holistic wellness, not just fitness performance

Key Insights Recap (from ANALYZE & SHARE)

From the analysis, the following insights were identified:

1. Most users fall into the moderate activity category (5,000–10,000 steps/day).
2. A strong positive relationship exists between steps and calories burned
3. Average sleep duration is approximately 7 hours, slightly below recommended levels.
4. Sleep duration varies by activity level, with highly active users sleeping marginally longer.
5. User engagement varies, indicating opportunities for personalized guidance.

Final Conclusion:

This analysis demonstrates that while smart device users are generally consistent in tracking physical activity, there are clear opportunities to improve engagement around **sleep quality and holistic wellness behaviours**. Most users fall within a moderate activity range and exhibit strong consistency in step tracking, but average sleep duration remains slightly below recommended levels. These patterns suggest that users benefit from guidance that goes beyond tracking and supports balanced, sustainable wellness habits.

Business Recommendations:

1. Personalized Wellness Coaching

Bellabeat should leverage user activity and sleep data to deliver personalized, actionable insights through the Bellabeat app. Rather than focusing solely on step counts, the platform can guide users toward incremental improvements based on their current activity level.

- Low-activity users can be encouraged with achievable daily goals
- Moderately active users can be guided toward consistency and balance
- Highly active users can be supported with recovery and sleep optimization insights

Business impact:

Personalized coaching can increase daily app engagement and position Bellabeat as an active wellness partner rather than a passive tracking tool.

2. Prioritize Sleep-Focused Features and Messaging

Sleep analysis revealed that average sleep duration is close to, but slightly below, recommended guidelines. Bellabeat has an opportunity to differentiate by placing greater emphasis on sleep consistency and recovery, which align strongly with its holistic wellness mission.

Recommended actions include:

- Sleep consistency reminders
- Insights that connect daytime activity with nighttime sleep outcomes
- Evening wellness prompts focused on relaxation and recovery

Business impact:

Enhancing sleep-focused features can deepen user trust, improve long-term retention, and reinforce Bellabeat's identity as a comprehensive wellness brand.

3. Implement Activity-Based User Segmentation for Marketing

Users demonstrate distinct behaviour patterns based on activity levels. Bellabeat can apply these insights to create **targeted marketing campaigns** that resonate with different user segments.

Examples include:

- Motivational messaging for low-activity users
- Habit-building and consistency-focused messaging for moderate users
- Balance and recovery-focused messaging for highly active users

Business impact:

Targeted messaging increases relevance, improves conversion rates, and reduces user fatigue caused by generic communications.

4. Use Content to Reinforce Data-Driven Wellness Education

Bellabeat can leverage blogs, in-app articles, and email content to educate users on the relationship between activity, sleep, and overall well-being. Content grounded in real usage data helps users understand *why* certain behaviors matter and encourages long-term habit formation.

Business impact:

Educational content strengthens brand credibility and encourages sustained engagement beyond device usage.

How Bellabeat Can Apply These Insights

- Improve user retention by offering **actionable, personalized insights**
 - Strengthen brand identity around **balanced wellness**
 - Drive higher engagement with underused features such as sleep tracking
 - Inform future product development and app enhancements
-

Limitations and Opportunities for Expansion:**Limitations**

- Small sample size (30 users)
- Short data collection period
- Incomplete sleep and weight tracking across users
- Lack of demographic context

Future Opportunities

- Analyze larger, more diverse user datasets
- Incorporate demographic and longitudinal data
- Test personalized notifications using A/B experiments
- Explore deeper links between sleep quality and long-term engagement

Executive Takeaway:

By applying data-driven insights from smart device usage, Bellabeat can move beyond basic activity tracking to deliver a more personalized, balanced, and impactful wellness experience. Focusing on sleep optimization, targeted engagement, and personalized coaching will allow Bellabeat to strengthen customer relationships, increase retention, and reinforce its position as a leader in holistic wellness technology.

Author's Comments:

I analysed Fitbit smart device data for a Bellabeat case study to identify trends in activity and sleep behaviour.

I cleaned and merged daily activity and sleep data in R, then analysed usage patterns. Personally, for me, Aligning and cleaning time-based data across multiple datasets to ensure accurate joins.

I found that most users are moderately active, with strong links between steps and calories, but sleep duration was slightly below recommended levels.

Based on this, I recommended personalized coaching, sleep-focused features, and targeted marketing to promote balanced wellness.
